

# Logic Dungeon II

## THE SYLLOGISM LABYRINTH

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**Purpose:** Use the logic tools from Weeks 4–6 to select the best-supported route through the labyrinth.

### The Second Seal

Beyond the Hall of Inscriptions, twenty adventurers reach a circular chamber. The next seal lies in the **Upper Archive**. Three routes leave the chamber:

- **Raven Stair** — an upward arrow is cut beneath an old raven seal.
- **Well Passage** — cold air flows through a round stone opening.
- **Lantern Gallery** — a blue survey thread disappears around its first turn.

No route may be accepted merely because it looks promising. At each station, the company must make a limited decision: record a clue, mark a route provisional, send a scout, avoid a trap, or repair a mistake.

**Two-test rule:** First apply the *form test*: if the premises were granted, would the conclusion follow? Then apply the *fact test*: are the premises established, doubtful, or false? A route can be best supported without being guaranteed safe.

**Tools:** premises and conclusion; major, minor, and middle terms; practical syllogism map; A/E/I/O mood; figure; undistributed middle; two negative premises; validity; premise pressure point; soundness.

*A perfect bridge drawn on rotten paper will not carry an expedition.*

## Part 1: The Raven Inscription

All routes bearing an Archive seal are routes that reach the Upper Archive.  
The Raven Stair is a route bearing an Archive seal.  
Therefore, the Raven Stair is a route that reaches the Upper Archive.

**Part 1 Q1:** List the major premise, minor premise, and conclusion. Name the major, minor, and middle terms. Identify the mood and figure, then apply the form test.

### Part 1 Q1 ANSWER BOX

**Part 1 Q2:** Identify the pressure-point premise and evidence that would test it. What may the company record now, and should Raven be accepted, rejected, or marked provisional?

### Part 1 Q2 ANSWER BOX

## Part 2: The Breath of the Well

A guard places a hand in the draft and argues:

All passages opening from the Upper Archive carry cold air.  
The Well Passage carries cold air.  
Therefore, the Well Passage opens from the Upper Archive.

**Part 2 Q1:** Let  $S$  = Well,  $P$  = passages opening from the Archive, and  $M$  = passages carrying cold air. Write the  $M/P/S$  pattern, identify mood and figure, and explain whether  $M$  connects  $S$  to  $P$ .

### Part 2 Q1 ANSWER BOX

**Part 2 Q2:** A scout can safely examine only the first turn. What observation may the company preserve, what has not been proved, and what limited investigation (if any) is justified?

### Part 2 Q2 ANSWER BOX

### Part 3: The Blue-Thread Rule

A dated card in the survey cabinet states:

**Record A — Chief Surveyor's Rule:** "Every route carrying an unbroken blue thread from this cabinet was inspected today and terminates in the Upper Archive." The signature and cabinet seal match today's log.

At Lantern, the company can see blue thread entering the gallery, but cannot yet tell which spool it came from or whether it remains unbroken.

**Part 3 Q1:** Map the proposed Lantern argument: All M are P; All S are M; therefore All S are P. Identify S, P, M, mood, figure, and the form-test result.

**Part 3 Q1 ANSWER BOX**

**Part 3 Q2:** Apply the fact test. Which premise is established, which remains doubtful, and what is the present diagnosis? Choose and justify one action: reject Lantern, accept it for all twenty, or send two scouts only far enough to inspect the thread.

**Part 3 Q2 ANSWER BOX**

## Part 4: The Fork of False Bridges

The two Lantern scouts reach a fork. One path descends toward standing water; the other rises beside a narrow ledge. Two warnings are carved there.

### Flooded-fork warning:

No flooded passages are dry passages.  
No ledge routes are flooded passages.  
Therefore, all ledge routes are dry passages.

**Part 4 Q1:** State the mood and test the form. Explain what two negative premises fail to connect. What trap decision should the scouts make about assuming the ledge is dry?

### Part 4 Q1 ANSWER BOX

### Rising-path warning:

All routes to the Upper Archive rise.  
This ledge route rises.  
Therefore, this ledge route goes to the Upper Archive.

**Part 4 Q2:** Identify the figure and form error. Explain why a true conclusion would still not make this good proof, and state the scouts' justified next action.

### Part 4 Q2 ANSWER BOX

## Part 5: Records That Change the Field

The scouts return safely after marking the fork *unproved*. Meanwhile, the main company recovers two current maintenance records.

**Record B — Mason's Ledger:** "Raven seals marked routes proposed for Archive access during the old construction. Some were completed; some were abandoned. The visible seals were never recut to show which."

**Record C — Cistern Report:** "The Well Passage crosses above an ice cistern. The cistern sends cold air into passages on both upper and lower levels." The airflow reading beside Well is current.

**Part 5 Q1:** How does Record B change Raven's fact test without changing its form? Give Raven's present diagnosis and the company's route consequence.

**Part 5 Q1 ANSWER BOX**

**Part 5 Q2:** What does Record C establish about Well, and why does it fail to repair the argument? Give Well's present diagnosis and route consequence.

**Part 5 Q2 ANSWER BOX**

## Part 6: The Recovered Spool

Near the false fork, a scout's marker cord catches a loose stone. The company uses a retreat line and recovers the scout without committing the other eighteen. Behind the stone is the missing spool ledger:

**Record D — Thread Check:** "Spool 3 runs from today's sealed survey cabinet into Lantern Gallery and remains unbroken through the first fork to the Upper Archive landing." Two surveyors signed the entry. The number stamped on the visible Lantern thread is 3.

**Part 6 Q1:** Rebuild the Lantern syllogism in standard order using Records A and D. Identify premises, conclusion, S/P/M, mood, figure, form judgment, and fact status.

### Part 6 Q1 ANSWER BOX

**Part 6 Q2:** What new decision is justified now that was not justified in Part 3? Distinguish reaching the Archive from safety, and propose a limited-risk movement for the twenty adventurers.

### Part 6 Q2 ANSWER BOX

**Part 7: Labyrinth Evidence Chart**

| Route   | Form | Pressure point | Fact status | Diagnosis |
|---------|------|----------------|-------------|-----------|
| Raven   |      |                |             |           |
| Well    |      |                |             |           |
| Lantern |      |                |             |           |

**Part 7 Q1:** Complete the chart. Then identify the valid argument with a doubtful premise, the invalid argument containing a true observation, and the sound argument. Explain why these differ.

**Part 7 Q1 ANSWER BOX**

**Part 7 Q2:** Which route is now best supported? Use at least three pieces of evidence from different parts and explain why each rejected route has weaker proof.

**Part 7 Q2 ANSWER BOX**



## Part 8: The Company's Decision

The stone guide declares: "Lantern's argument is valid. Therefore its premises must be true, and the route must be safe." The company pauses before moving all twenty adventurers.

**Part 8 Q1:** Separate and repair the guide's claims. State what the valid form establishes only when combined with the records, what is best supported, and what remains unguaranteed.

### Part 8 Q1 ANSWER BOX

**Part 8 Q2:** Give the company's final movement plan. Include one observation or safety check, one repair or retreat provision, and a 2–3 sentence explanation of why both the form test and fact test were needed.

### Part 8 Q2 ANSWER BOX

*The company follows the evidence, not the loudest sign. The second seal waits above.*